EX NO: 1	Estimation of Characteristic Impedance (Zo), and Propagation Constant (γ) of a Transmission Line from Primary Constants
DATE: 08-10-2026	

Test Case 1:

Investigating the significance of Characteristic Impedance (Zo), and Propagation Constant (γ) of a Transmission Line using the following parameters

a) Resistance =10 Ω/Km

b) Capacitance = 8 μF/Km

c) Inductance =3.5 mH/Km

d) Conductance =4 μmho/Km

e) Frequency= 2 Khz

AIM: Investigating the significance of Characteristic Impedance (Zo), and Propagation Constant (γ) of a Transmission Line with Resistance =10 Ω/Km, Capacitance = 8 μF/Km, Inductance =3.5 mH/Km, Conductance =4 μmho/Km, Frequency= 2 Khz

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clear; clc;

R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);

w=2*%pi*f;

Z=R+(%i*w*L);

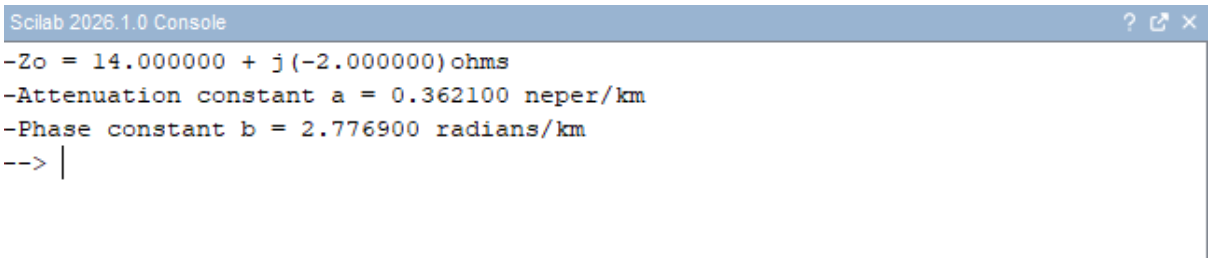
Y=G+(%i*w*C);

Zo=sqrt(Z/Y);

C=round(real(Zo));
```

```
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

OUTPUT:



Constants L=3mH; f=2000; G=.4*(10^-6); C=8*(10^-6).

Simulation Results			Theoretical Results	
Z0(ohms)	α(neper/km)	β(radians/km)	α(neper/km)	β(radians/km)
14.000000 + j(-2.000000)ohms	0.362100 neper/km	2.776900 radians/km		

RESULT:

The characteristic impedance and propagation constant of a transmission line is calculated using SCILAB and the impact of primary constants on secondary constants are investigated.

Test Case 2:

Examine the impact of resistance and inductance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line at a frequency f=2.5 Khz. Vary the resistance value from 10 Ω to 50 Ω at the interval of 10 Ω with L,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the inductance values from 3.5 mH to 11.5 mH at the interval of 2.5 mH with R,G and C values constant.

AIM: To examine the impact of resistance and inductance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line at a frequency f=2.5 Khz. Vary the resistance value from 10 Ω to 50 Ω at the interval of 10 Ω with L,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the inductance values from 3.5 mH to 11.5 mH at the interval of 2.5 mH with R,G and C values constant.

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clear; clc;

R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);

w=2*%pi*f;

Z=R+(%i*w*L);

Y=G+(%i*w*C);

Zo=sqrt(Z/Y);

C=round(real(Zo));

D=round(imag(Zo));

printf('-Zo = %f + j(%f)ohms\n',C,D);

P=sqrt(Z*Y);
```

```
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

OUTPUT:

```
Scilab 2026.1.0 Console
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
--> exec('D:\TRANSMISSION\EXP 1 TC2.sci', -1)
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
--> exec('D:\TRANSMISSION\EXP 1 TC2.sci', -1)
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
--> exec('D:\TRANSMISSION\EXP 1 TC2.sci', -1)
```

Variable L

Constants R=10Ω; f=2000; G=.4*(10^-6); C=8*(10^-6).

S. No.	Inputs (L)	Simulation Results			Theoretical Results	
		Z0(ohms)	α (neper/km)	β (radians/km)	α (neper/km)	β (radians/km)
1)	0.003	14.000000 + j(-2.000000)ohms	0.362100 neper/km	2.776900 radians/km		
2)	0.004	16.000000 + j(-2.000000)ohms	0.314700 neper/km	3.194600 radians/km		
3)	0.005	18.000000 + j(-1.000000)ohms	0.282000 neper/km	3.565500 radians/km		
4)	0.009	24.000000 + j(-1.000000)ohms	0.210700 neper/km	4.773200 radians/km		

5)	0.012	27.000000 + j(-1.000000)ohms	0.182500 neper/km	5.509300 radians/km		

RESULT:

The impact of resistance and inductance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line is examined at a frequency f=2.5 Khz.

Test Case 3:

Explore the impact of frequency and capacitance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the capacitance values from 8 µF to 16 µF at the interval of 2 µF with R,G and L values constant.

AIM: To explore the impact of frequency and capacitance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant

PROGRAM:

```
clear; clc;

R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);

w=2*%pi*f;

Z=R+(%i*w*L);

Y=G+(%i*w*C);

Zo=sqrt(Z/Y);

C=round(real(Zo));

D=round(imag(Zo));

printf('-Zo = %f + j(%f)ohms\n',C,D);

P=sqrt(Z*Y);

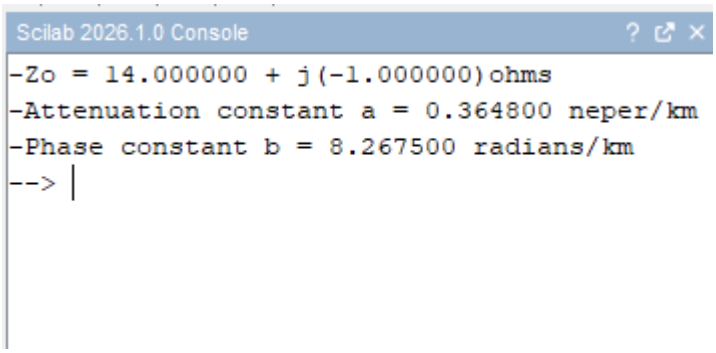
a=real(P);

a1=round(a*10000)/10000;

printf('-Attenuation constant a = %f neper/km\n',a1);
```

```
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

OUTPUT:



Variable f

Constants L=3mH; f=2000; G=.4*(10^-6); C=8*(10^-6).

S. No.	Inputs (f) Hz	Simulation Results			Theoretical Results	
		Z0(ohms)	α (neper /km)	β (radians/ km)	α (neper/km)	β (radians/k m)
1)	2000	14.000000 + j(-2.000000)ohms	0.362100 neper/km	2.776900 radians/km		
2)	3000	14.000000 + j(-1.000000)ohms	0.363800 neper/km	4.145700 radians/km		
3)	4500	14.000000 + j(-1.000000)ohms	0.364500 neper/km	6.205300 radians/km		

4)	5500	14.000000 + j(-1.000000)ohms	0.364800 neper/km	7.580000 radians/km		
5)	6000	14.000000 + j(-1.000000)ohms	0.364800 neper/km	8.267500 radians/km		

RESULT:

The impact of frequency and capacitance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line is investigated for various values of frequencies and capacitance values.